A2 past final worksheet

1. (Fall 2024, #2) Find all values of k and a so that

$$f(x) = \begin{cases} \dfrac{3\sin(kx^2)}{x}, & x > 0 \\ 6 - 3k + ax, & x \leq 0 \end{cases}$$

is continuous and differentiable at $x = 0$.

2. (Fall 2023, #2) Consider the piecewise function

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x^2} + 3 & \text{for} \quad x \neq 0 \\ 3 & \text{for} \quad x = 0 \end{cases}$$

(a) Is f continuous at $x = 0$? Justify.

(b) Is f differentiable at $x = 0$? (Justify). If it is, find the value of $f'(0)$.

3. (Spring 2023, #2) The gravitational force, F, exerted by a planet of (constant) radius R on an object a distance x away from the center of the planet is given by the function

$$F(x) = \begin{cases} \dfrac{kx}{R^3} & \text{if } x \leq R \\ \\ \dfrac{k}{x^2} & \text{if } x > R \end{cases}$$

where k is some constant.

(a) Use the definition of continuity to determine if $F(x)$ is continuous at R.

(b) Use the limit definition of the derivative to find an expression for $F'(x)$ when $x > R$.

4. (Spring 2019, #3) Find the derivative of the following function **using the definition**. No credit will be given for using derivative rules.

$$f(x) = \frac{1}{1+x^2}$$

5. (Fall 2018, #4) Let $f(x) = \begin{cases} x^2 \cos\left(\dfrac{\pi}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$

(a) Show that $f(x)$ is continuous at $x = 0$.

(b) Is $f(x)$ differentiable at $x = 0$?